

Database Technologies – Assignment 09

1. Create an index that will enable a user to pull orders for '1990-10-03' out of the Orders table quickly.

```
D6_96342_Jagdish@localhost> CREATE INDEX idx_ord_date ON orders(odate);
Query OK, 0 rows affected (0.014 sec)
Records: 0 Duplicates: 0 Warnings: 0

D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE odate = "1990-10-03";
+-----+-----+-----+-----+-----+
| onum | amt  | odate | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
5 rows in set (0.001 sec)

D6_96342_Jagdish@localhost> EXPLAIN ANALYZE SELECT * FROM orders WHERE odate = "1990-10-03";
+-----+-----+-----+-----+-----+
| EXPLAIN
+-----+-----+-----+-----+-----+
| -> Index lookup on orders using idx_ord_date (odate = DATE'1990-10-03') (cost=1 rows=5) (actual time=0.0316..0.0413 rows=5 loops=1)
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
D6_96342_Jagdish@localhost> CREATE UNIQUE INDEX idx_uni_onum ON orders(onum);
Query OK, 0 rows affected (0.014 sec)
Records: 0 Duplicates: 0 Warnings: 0

D6_96342_Jagdish@localhost> INSERT INTO orders VALUES(3001, 500, "1990-10-03", 2009, 1008);
ERROR 1062 (23000): Duplicate entry '3001' for key 'orders.idx_uni_onum'
D6_96342_Jagdish@localhost> █
```

3. Create an index that would permit salesperson to retrieve his orders.

```
D6_96342_Jagdish@localhost> CREATE INDEX idx_sp_orders ON orders(snum);
Query OK, 0 rows affected, 1 warning (0.019 sec)
Records: 0 Duplicates: 0 Warnings: 1

D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE snum = 1001;
+-----+-----+-----+-----+-----+
| onum | amt  | odate | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)

D6_96342_Jagdish@localhost> EXPLAIN ANALYZE SELECT * FROM orders WHERE snum = 1001;
+-----+-----+-----+-----+-----+
| EXPLAIN |
+-----+-----+-----+-----+-----+
| -> Index lookup on orders using idx_snum (snum = 1001) (cost=0.8 rows=3) (actual time=0.0353..0.0462 rows=3 loops=1) |
| |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
D6_96342_Jagdish@localhost> CREATE UNIQUE INDEX idx_uni_cust_rating ON customers(rating, snum);
Query OK, 0 rows affected (0.015 sec)
Records: 0 Duplicates: 0 Warnings: 0

D6_96342_Jagdish@localhost> INSERT INTO customers VALUES(2009, "James", "Pune", 200, 2001);
Query OK, 1 row affected (0.002 sec)

D6_96342_Jagdish@localhost> INSERT INTO customers VALUES(2009, "James", "Pune", 200, 1001);
ERROR 1062 (23000): Duplicate entry '200-1001' for key 'customers.idx_uni_cust_rating'
```

5. Create an index to speed up searching order on a given date by given customer.

```
D6_96342_Jagdish@localhost> CREATE INDEX idx_date_cust ON orders(odate, cnum);
Query OK, 0 rows affected (0.014 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE odate = "1990-10-03" AND cnum = 2001;
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001

1 row in set (0.001 sec)

```
D6_96342_Jagdish@localhost> EXPLAIN ANALYZE SELECT * FROM orders WHERE odate = "1990-10-03" AND cnum = 2001;
```

EXPLAIN
-> Index lookup on orders using idx_date_cust (odate = DATE'1990-10-03', cnum = 2001) (cost=0.35 rows=1) (actual time=0.0379..0.044 rows=1 loops=1)

1 row in set (0.001 sec)